RealSampleCalibration — Beamline Staff Guide

ANTARES beamline (SOLEIL) · calibrating the stage model with RealSampleCalibration, and looking after the program

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What the staff are responsible for

Users only *read* the stage calibration: QuickAlignment loads the newest **configuration file** for the chosen optic and has no way to change it. The configuration is made by beamline staff with the separate program **RealSampleCalibration** and stored in `configs/`:

```
configs/calibration_YYYY-MM-DD_HHMMSS_<optic>.json      optic: zone_plate or capillary
```

One file per calibration, never overwritten. What remains for a user (mostly the error of drawing a sample on the plate) is corrected by *Found here* in the Stage & Optics tab.

The stage model and its constants

Stack, bottom to top: ST / SN / SZ translations → SRz → SRn → PIX / PIY piezo → plate. Top view: the beam runs along +y, the analyzer focal plane at 45°, ST along it and SN perpendicular to it.

constant	meaning	determined by
ST0, SZ0	ST / SZ putting the plate centre on the beam (SRn 120°, SRz 0°)	set 1
SN_f	SN putting the standard plate's raised surface (5.0 mm) in the analyzer focal plane	set 1
SRn axis x, y	position of the SRn rotation axis on the plate	a set with SRn rotated
SRz offset ST, SN	offset of the SRz axis from the plate-surface centre line	a set with SRz rotated
optic reference	optic position putting the focus on the analyzer focus	optic position noted in the sets

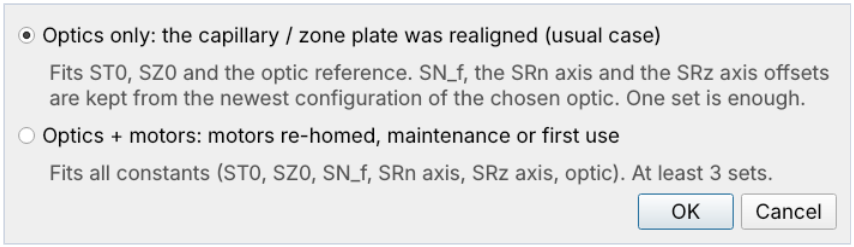
SRn = 120° (plate vertical, tab up) is the mechanical reference; PIX = PIY = 150 µm during calibration. The optic is modelled as a focus point moving along the beam: ST, SZ and SRn do not change the surface plane, so they need no optic move; SN, SRz and the sample height do (1 mm along SN moves the spot √2 mm along the beam).

When to recalibrate

situation	mode
capillary or zone plate realigned (beam meets the plate elsewhere, focus moved)	**optics only**
motors re-homed / reset, manipulator maintenance, receptacle changed	**optics + motors**
first use, or no configuration for this optic yet	**optics + motors**
users' *Found here* corrections above ~0.3 mm, same shift for several samples	check with optics only; full if the residuals stay large

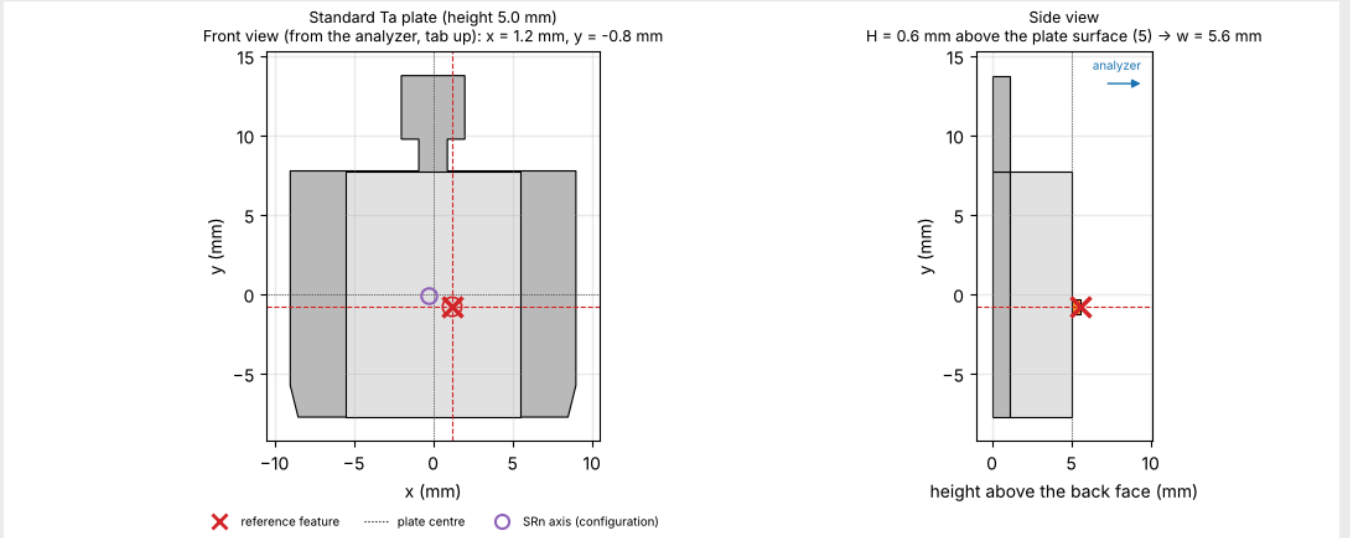
Starting RealSampleCalibration

Double-click **`RealSampleCalibration.py`** (Windows: `RealSampleCalibration.pyw`). The first question is what was recalibrated:



Optics only is offered only when a configuration exists already (the mechanics are taken from it).

In both modes, first enter the **reference feature**: one small, sharp feature that can be found with the beam (the corner of a reference sample, a mark), close to the plate centre so that it stays reachable when rotating. Enter its plate coordinates x, y (mm, from the centre of the back plate, front view, tab up) and its height H above the plate surface, and choose the **focusing optic**. **Show front / side view** draws the plate with the feature and lists where each set should find it with the current configuration (motor ranges checked):



Expected positions with the current fit:

set	SRn	SRz	ST	SZ	SN
1 reference	120	0	-2.109	41.400	0.074
2 SRn rotated	140	0	-2.275	40.858	0.074
3 SRz rotated	120	-10	-2.640	41.400	-0.221
4 SRn rotated (check)	100	0	-1.768	41.853	0.074
5 SRz rotated (check)	120	10	-1.535	41.400	0.272

Close

Use the **same feature in every set**.

Optics only (the usual case)

Optics recalibration – measurement protocol

When: the focusing optic (capillary or zone plate + OSA) was realigned, the motors were not re-homed. The beam then meets the plate elsewhere and the optic focus moves, but the mechanics stay.

What is fitted: ST0, SZ0 (where the beam meets the plate) and the optic position at the analyzer focus. **Kept** from the newest configuration of the chosen optic: SN_f, the SRn axis and the SRz axis offsets.

Reference feature: one small, sharp feature that you can find with the beam, with known plate coordinates x, y (mm, from the centre of the back plate, front view, tab up) and height H above the plate surface. Check it with *Show front / side view*. Use **the same feature in every set**.

For every set (as the stage is normally run):

1. Set SRn and SRz to the values of the set; PIX = PIY = 150 µm.
2. Leave SN as it is (it does not have to be at the analyzer focus).
3. Bring the feature onto the beam with ST / SZ.
4. Refocus the optic on the feature.
5. Write down ST, SZ, SN and the optic position.

Sets: set 1 (SRn 120, SRz 0) is enough. Sets 2–3 (SRn and SRz rotated) are recommended: they check the kept axes. If their residuals stay above ~0.05 mm the mechanics have changed (motors re-homed?): then start the program again and choose *optics + motors*.

Then press *Compute*, work through the test points (keep SRn, SRz and SN; enter the real ST, SZ and optic if the feature is not found within 0.05 mm), add a comment and press *Save*.

Reference feature (plate coordinates)

Plate: Standard Ta plate (height 5.0 mm)
x: 1.200 mm
y: -0.800 mm
H above plate surface: 0.600 mm
Show front / side view

Optic

Focusing optic: Capillary
Optic motor: + moves the optic towards the sample.
Mechanics kept from the newest configuration **2026-10-24 23:55:00 capillary**: SN_f -0.5250, SRn axis x -0.2592, SRn axis y -0.0493, SRz offset ST -0.2265, SRz offset SN 2.5869

Measurement sets: reference feature on the beam, optic refocused, SN as it was

Use	Set	SRn (°)	SRz (°)	ST (mm)	SZ (mm)	SN (mm)	Optic at focus
1 ☑	1 reference	120	0				
2 ☑	2 SRn rotated (check)	140	0				
3 ☑	3 SRz rotated (check)	120	-20				

Add set | Remove selected set

Compute

Fit results | Test points

Fill in the measured ST, SZ, SN and press Compute.

comment, e.g. 'after motor re-homing, 2026-10'

Save as new configuration

Configuration folder: QuickAlignment/configs

Fitted: ST0, SZ0 and the optic reference. Kept from the newest configuration of the chosen optic: SN_f, the SRn axis and the SRz axis offsets. For every set, as the stage is normally run:

1. Set SRn and SRz of the set; PIX = PIY = 150 µm.
2. **Leave SN as it is** (it does not have to be at the analyzer focus).
3. Bring the feature onto the beam with ST / SZ.
4. Refocus the optic on the feature.
5. Write down ST, SZ, SN and the optic position.

Set 1 (SRn 120°, SRz 0°) is enough. Sets 2–3 (SRn and SRz rotated) are recommended: they check the kept axes. If their residuals stay above ~0.05 mm the mechanics have changed: restart the program and choose *optics + motors*.

Optics + motors (full calibration)

All 7 constants and the optic reference are fitted. The feature is put **at the analyzer focus** in every set.

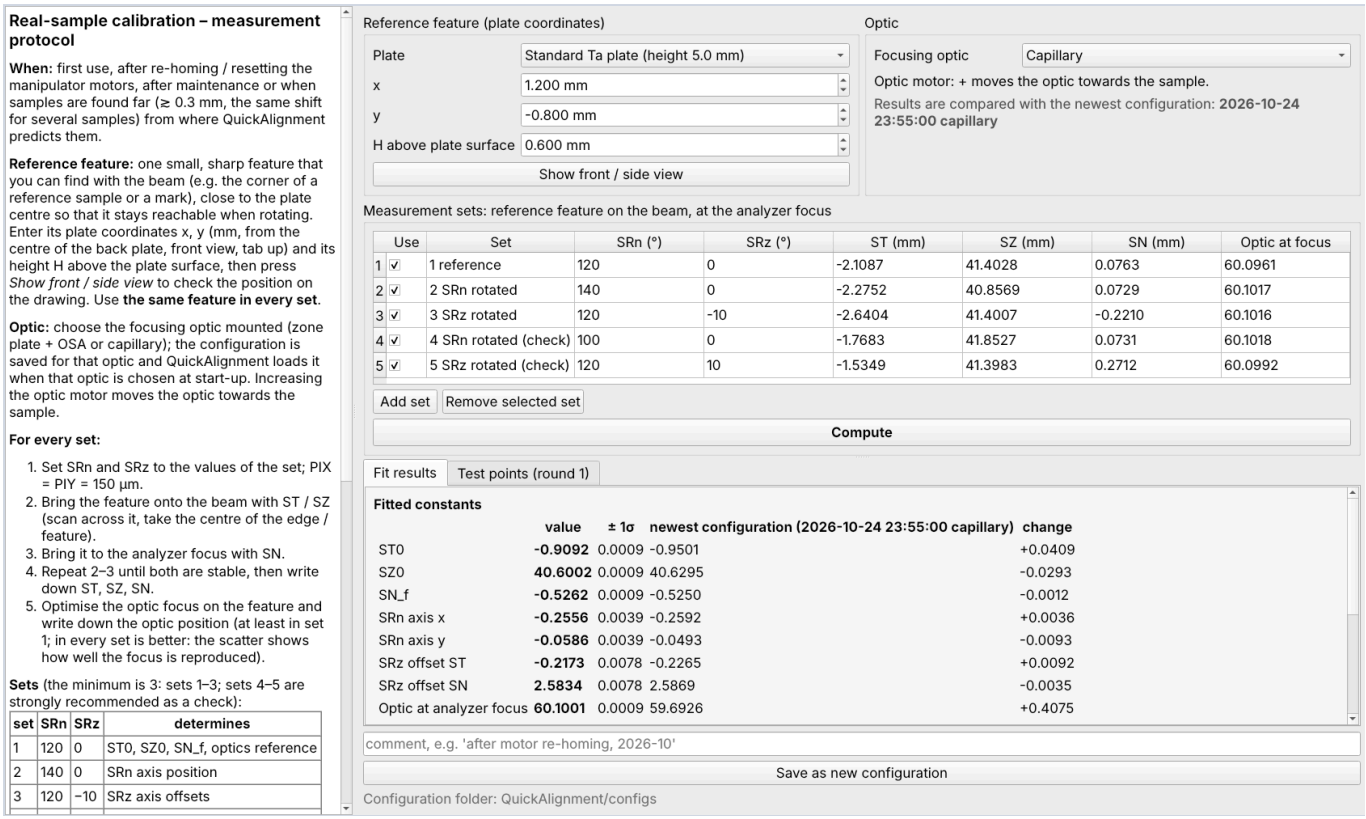
set	SRn	SRz	purpose
1	120	0	ST0, SZ0, SN_f, optic reference
2	140	0	SRn axis
3	120	−10	SRz axis
4	100	0	check (recommended)
5	120	+10	check (recommended)

Minimum 3 sets (each gives ST, SZ, SN); **5 recommended**, so that a wrong measurement shows up in the residuals. Larger rotations give more precise axes (|SRn − 120| ≥ 20°, |SRz| ≥ 10° as far as the geometry allows). Untick a set that was not measured; *Add set* adds one at other angles.

For every set:

1. Set SRn and SRz; PIX = PIY = 150 µm.
2. Bring the feature onto the beam with ST / SZ (scan across it, take the centre of the edge or feature).
3. Bring it to the analyzer focus with SN.
4. Repeat 2–3 until stable; write down ST, SZ, SN.
5. Optimise the optic focus on the feature and write down the optic position: at least in set 1, preferably in every set (+ moves the optic towards the sample).

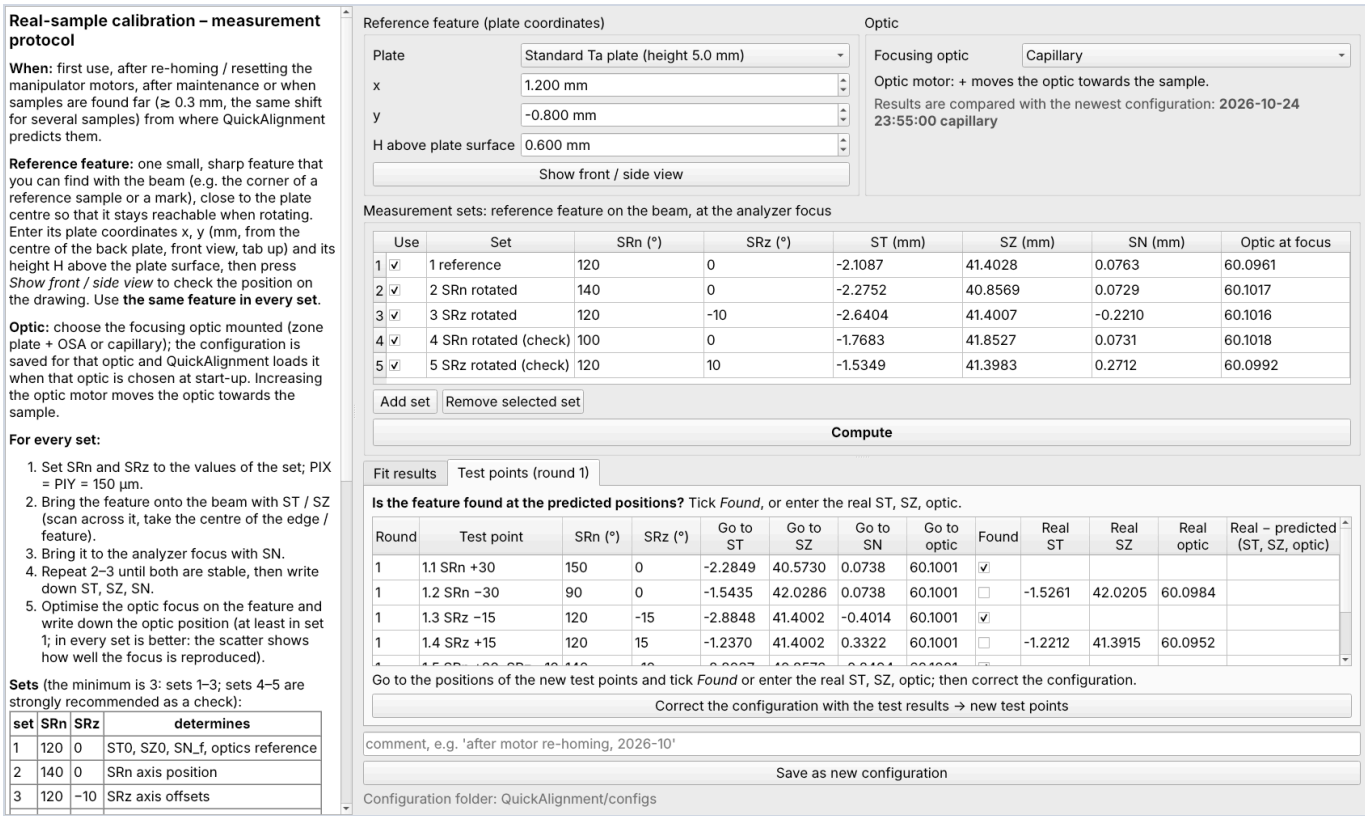
Press **Compute**:



- **Fitted constants** with their 1σ uncertainty, the value of the newest configuration and the change.
- **Residuals** per set (model − measured, mm) for ST, SZ, SN and the optic. With 5 sets they should be ≲ 0.01–0.02 mm. One large residual usually means that set was measured on another point or out of focus: re-measure it, or untick it.
- Warnings: *Not enough data* (which constants the sets cannot determine), *minimum data set* (mistakes cannot be detected), residuals above 0.05 mm, no optic position entered (optic reference not calibrated).
- A fit on impossible numbers (typing errors, NaN) is reported instead of being saved.

Test points: checking and correcting

After *Compute*, the **Test points** tab predicts rotations not used in the sets (SRn ± 30°, SRz ± 15°, SRn and SRz combined) with the ST, SZ, SN and optic position where the configuration puts the feature:



For each test point:

1. Go to the predicted SRn, SRz, ST, SZ, SN and optic position.
2. If the feature is on the beam and in focus **within 0.05 mm**, tick **Found**.
3. If not: keep SRn, SRz and SN, bring the feature onto the beam with ST / SZ, focus the optic and enter the **real ST, SZ and optic position** (optic optional). Red predictions are outside a motor range: change the angles of that test point (it is predicted again).

Correct the configuration with the test results → new test points adds the real positions to the fit (found points only confirm), fits all constants again and predicts a new round. Answered rounds stay in the table, read-only, with their deviations. Repeat until all test points of a round are found, or the deviations are small enough; a set that stays an outlier was probably measured wrongly: untick it and correct again.

Saving and how QuickAlignment uses the file

Add a comment (date, reason, what was measured) and press **Save as new configuration**. The file is written atomically (a failed write leaves no partial file) and contains the constants, the optics reference, the sets, the test points and their residuals, for the record.

- QuickAlignment loads, at every start, the **newest file of the optic chosen** by the user. A zone-plate configuration does not replace the capillary one and vice versa.
- If there is no file for that optic, the stage constants of the newest file are used and the optic position is shown as uncalibrated.
- Damaged or foreign files in `configs/` are skipped. To withdraw a bad configuration, move its file out of `configs/` (do not edit it).
- Users can choose an older configuration for their session in *Configuration...* (read-only).
- The motor limits of ST, SZ, SN, SRn and SRz can be changed in *Configuration...* → *Motor limits*; they are saved in `configs/motor_limits.json` (delete the file to go back to the built-in limits).

Anti-collision solids

The analyzer nose, the zone plate and OSA and the capillary are modelled as simple solids (QuickAlignment/README.md, *Anti-collision*). The zone plate / OSA dimensions are **placeholders** (the capillary working distance, 7.2 mm, is the beamline value): measure them (distance of each front from the focus, width, thickness; for the OSA, the OSA_S position they belong to) and put them in the configuration under `"collision"` (edit a copy, then save it as a new file). RealSampleCalibration keeps the solids of the previous configuration of the same optic.

Looking after the program

Installation (Python ≥ 3.9): `pip install -r requirements.txt` (PyQt5 or PySide6, numpy, matplotlib; pytest for the tests). Start with `QuickAlignment/QuickAlignment.py` / `RealSampleCalibration/RealSampleCalibration.py` (`.pyw` on Windows without console).

Tests: `python -m pytest` in the repository folder (about 200 tests, ~2 min; Qt runs off-screen). They cover the stage model, rotations, calibration fits, file loading and the main windows.

Limits that may need editing (one place each):

what	where
motor travel ranges ST, SN, SZ (input fields and warnings)	`MOTOR_SPECS` in `src/quickalignment/core/motors.py`
usable SRn / SRz, deflector limits, accepted offsets, photon energy range, zone-plate energy, work function	constants at the top of `src/quickalignment/alignment/bz_target.py`
analyzer slit orientation and the signs of slit / deflector angles	`AnalyzerGeometry` in `src/quickalignment/alignment/rotation.py`
plate drawings (standard, gating)	`src/quickalignment/core/geometry.py`

Robustness: an unexpected error in the program is shown in a dialog (with details) and the program keeps running, so that entered positions are not lost; please collect these details. Sessions and configurations are written atomically; unreadable sessions give a message instead of a crash; impossible table entries (nan, inf, text) are refused.

Guides: the screenshots are made by `python devtools/guide_screenshots.py`, the PDFs by `python devtools/build_guides.py` (needs pandoc and Playwright with Chromium). Sources: `QuickAlignment/docs/USER_GUIDE_*.md`, `RealSampleCalibration/docs/STAFF_GUIDE_*.md`.

Troubleshooting

symptom	likely cause / action
all samples found shifted by the same amount	ST0 / SZ0 changed (optic realigned): *optics only*
shift depends on the rotation	axes changed (motors re-homed): *optics + motors*
Optics only residuals of sets 2–3 above 0.05 mm	mechanics changed: *optics + motors*
one set with a large residual	wrong point, out of focus or typing error: re-measure or untick

symptom	likely cause / action
Not enough data: …	the listed constants need a set with SRn and/or SRz rotated
a set needs SN below the collision limit	the feature is too low; use a feature on a higher sample
QuickAlignment shows *default* configuration	no readable file in `configs/` for any optic